



COLLEGE CODE: 9528

COLLEGE NAME: SCAD COLLEGE OF ENGINEERING AND
TECHNOLOGY

DEPARTMENT: COMPUTER SCIENCE AND
ENGINEERING

STUDENT NMID: 5BA78A6BB207CF6931040AE332BD179E

Roll no : 952823104149

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Phase 2

TECHNOLOGY PROJECT NAME :

E-Commerce Product Page

SUBMITTED By,

NAME: SHALINI .I

MOBILE NO: 9894079303

Phase2- Solution Design &architecture

1.Tech Stack

Selection

To ensure scalability, performance, and ease of maintenance, the following technologies are chosen:

- Frontend → React.js
 - Provides a fast, responsive, and component-based UI for the product page.
- Backend → Node.js with Express
 - Used to build REST APIs for products, cart, wishlist, and reviews.
- Database → MongoDB
 - A NoSQL database with a flexible schema, suitable for handling products, user carts, and reviews.
- Authentication → JWT (JSON Web Tokens)
 - Ensures secure login and session management for users.
- Deployment → Vercel for frontend hosting and AWS/Heroku for backend services.

2. UI Structure / API Schema Design

UI Structure (Main Components):

- ProductPage → Parent container for the page.
- ProductImageGallery → Displays product images in a slider.
- ProductInfo → Shows product title, description, price, and stock status.
- AddToCartButton → Lets users add products to cart.
- WishlistButton → Saves product for later.
- RatingsAndReviews → Shows average rating and customer reviews.
- RelatedProducts → Suggests similar products.

Sample Product API Schema:

```
{
  "productId": "P12345",
  "name": "Wireless Bluetooth Headphones",
  "price": 1999,
  "description": "High-quality wireless headphones with noise cancellation.",
  "images": ["img1.jpg", "img2.jpg"],
  "stock": 12,
  "category": "Electronics",
  "rating": 4.3,
  "reviews": [
    {
      "user": "John",
      "rating": 5,
      "comment": "Excellent sound quality!"
    },
    {
      "user": "Priya",
      "rating": 4,
      "comment": "Battery life could be better."
    }
  ]
}
```

Key API Endpoints:

- GET /api/products/{id} → Fetch product details
- POST /api/cart/add → Add product to cart
- POST /api/wishlist/add → Add product to wishlist
- GET /api/reviews/{productId} → Fetch reviews
- POST /api/reviews/{productId} → Add new review

3.Data Handling Approach

To ensure smooth interaction between frontend, backend, and database, the following approach is used:

- Frontend → Uses Axios/React Query for API calls with caching, loading states, and error handling.
- Backend → Validates product ID, stock availability, and price before updating the database.
- Database →
 - products → Stores product details
 - cart → Stores user cart items
 - reviews → Stores customer feedback
- Error Handling →
 - 200 → Success
 - 400 → Invalid Request
 - 404 → Product Not Found
 - 500 → Server Error

4.component/Module diagram

Frontend Components

- ProductPage.js
- ProductImageGallery.js
- ProductInfo.js
- AddToCart.js
- WishlistButton.js
- ReviewsSection.js

Backend Modules

- ProductController.js → Handles product-related APIs
- CartController.js → Manages cart operations
- ReviewController.js → Manages review operations
- AuthMiddleware.js → Handles authentication & authorization

5.Basic Flow Diagram

- User visits Product Page
- System calls GET /api/products/{id} → Fetch product details from DB
- UI displays product info, images, price, stock, and reviews
- User clicks Add to Cart → API POST /api/cart/add is triggered
- Backend validates product availability → Updates cart in DB
- API responds with success → Cart count updates on UI
- User can also add product to wishlist or submit reviews